

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2021

ESE 4025

(HVAC&R System Design)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

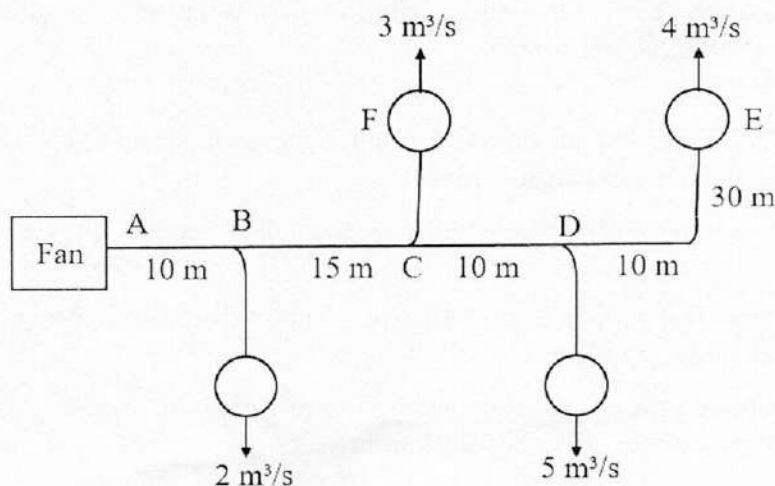
iv) Required tables and charts will be provided on request.

SECTION – A

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|-------|--|----|
| 1(a). | Define Ton and COP of refrigeration system. What is the basic difference between a refrigerator and a heat pump? | 08 |
| 1(b). | How does the increase in condenser temperature affect on COP? Also briefly explain the influence of evaporator temperature on COP. Which of the two temperatures have more influence on COP? | 12 |
| 1(c). | A refrigeration system using R22 is to have a refrigerating capacity of 80 kW. The system follows vapor compression cycle with 5°C subcooling and 10°C superheating in which the evaporating and condensing temperature are –10°C and 45°C, respectively. Determine:
(i) the volume flow rate of refrigerant
(ii) COP
(iii) the power required to drive the compressor. | 15 |
| 2(a). | What is meant by multipressure systems? Discuss about the individual compressors-multiple expansion valves with the required diagrams. | 12 |
| 2(b). | Briefly discuss the desirable characteristics of a fluid to be used as a refrigerant. | 08 |
| 2(c). | An ammonia refrigerator produces 30 tonnes of ice from and at 0°C in 24 hours. The temperature range of the compressor is from 20°C to –10°C. The vapor is dry saturated at the end of compression and an expansion valve is used. Assume a COP to 55% of the theoretical value. Calculate the power required to drive the compressor. Latent heat of ice is 335 kJ/kg. | 15 |
| 3(a). | What are the different methods of air refrigeration systems? Explain the simple air cooling system with necessary diagrams. | 10 |
| 3(b). | Differentiate between thermo-electric and electro-caloric cooling systems. | 05 |
| 3(c). | Describe the Linde system of liquifaction of air. | 10 |
| 3(d). | What is meant by hermetic sealed compressor? What are the advantages and disadvantages of this compressor? | 10 |
| 4(a). | What is the function of expansion valve in a refrigeration system? Explain the working principle of thermostatic expansion valve. | 12 |
| 4(b). | What are the advantages of vapor absorption system over vapor compression refrigeration system? | 06 |
| 4(c). | What are the practical problems in LiBr-water vapor absorption system? Write a short note on crystallization problem. | 07 |
| 4(d). | Draw the schematic of a single effect absorption refrigeration system. Also explain how it is different from double effect absorption refrigeration system. | 10 |

SECTION – B

- 5(a). Define the following terms: 12
- (i) Dew point temperature (ii) Relative humidity
- (iii) Wet bulb temperature (iv) Sensible heat factor.
- 5(b). What is psychrometry? Describe the following process in a psychrometric chart: 08
- (i) Cooling and dehumidification
- (ii) Heating and humidification.
- 5(c). The room sensible heat gain and latent heat gain of a conditioned space are 40 kW and 15 kW respectively. The inside design conditions are 25° DBT and 55% RH. The return air is mixed with outside air before entering the cooling coil in the ratio of 3:1 by weight. Assuming the bypass factor of the cooling coil is 0.13, determine: 15
- (i) Apparatus dew point (ADP)
- (ii) Ventilation air in cmm
- (iii) Total refrigeration load of the plant.
- 6(a). What is meant by thermal comfort? What are the factors affecting thermal comfort? 08
- 6(b). What is meant by zone? Explain the difference between constant air volume and variable air volume air-conditioning system with schematic diagram. 09
- 6(c). What are the advantages and disadvantages of rooftop air-conditioning unit? 08
- 6(d). Write short notes on: (i) Filtration, and (ii) Dilution. 10
- 7(a). What is meant by VRF system? Explain three-pipe heat recovery VRF systems with required sketches. 12
- 7(b). What are the differences between a heat pump multisplit system and a VRF multisplit system? 08
- 7(c). What is meant by cooling load? What assumptions behind cooling load design? 07
- 7(d). What is sol-air temperature? How to estimate the solar heat gain through building window? 08
- 8(a). What is infiltration? How to estimate infiltration rate using Crack method? 07
- 8(b). What factors should be considered to locating the supply air outlets and return air inlets? 08
- 8(c). What are the general rules for duct design? 06
- 8(d). For the duct system as shown in figure, determine the dimensions AB, BC, CD and DE of the rectangular ducts with aspect ratio 3:1 assuming friction rate of 0.1 mm H₂O/m length duct. 14



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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2021

ESE 4115

(Power Plant Engineering)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
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SECTION – A

- 1(a). What is meant by power plant engineering? What are the basic sources of energy widely used in power generation. 06
- 1(b). Describe a typical daily load curve of Bangladesh. 08
- 1(c). Define the following terms: 09
- (i) Plant capacity factor
 - (ii) Diversity factor
 - (iii) Demand factor

- 1(d). A power station has the following daily load cycle: 12

Time in Hours	6 – 8	8 – 12	12 – 16	16 – 20	20 – 24	24 – 6
Load in MW	15	35	65	25	60	20

Draw the load curve and load duration curve. Also calculate the energy generated per day.

- 2(a). What are the effects of variable load on power plant? 06
- 2(b). What are the factors to be considered while selecting site for a power plant? 07
- 2(c). What are the benefits of interconnected grid system? Describe. 10
- 2(d). A power plant of 100 MW installed capacity has the following particulars: 12
- Capital cost = Tk. 25,000 per kW installed
 - Interest and depreciation = 10%.
 - Annual load factor = 50%
 - Annual capacity factor = 45%
 - Annual running charge = Tk. 500×10^6
 - Energy consumed by power plant auxiliaries = 5%

Calculate the cost of generation per kWh

- 3(a). What is load forecasting? What are factors affecting load forecasting? 10
- 3(b). Distinguish between peak load approach and energy approach of load forecasting. 06
- 3(c). What is meant by tariff? Write down the objectives and requirements of a tariff. 12
- 3(d). Explain the future power demand of Bangladesh according to the power system master plan 2015. 07

4(a).	Describe the essential features of a hydro-electric power plant with neat sketch. Write the functions of surge tank and forebay.	09
4(b).	Explain the governing of impulse turbine with neat sketch.	07
4(c).	Define hydrograph and mass curve.	04
4(d).	The catchment area for the reservoir of a hydro-electric power plant is 14×10^8 sq.m having an average rainfall per year of 125 cm, 70% of the rainfall is available for power generation and the mean operating head is 40 m. Calculate the maximum plant capacity to be installed. Assume that 5% of the head is lost in penstock, turbine efficiency 85%, generator efficiency 90% and annual load factor is 45%.	15

SECTION – B

5(a).	Describe the working principle of diesel power plant with the help of schematic diagram.	20
5(b).	Write down the importance of lubrication system in diesel power plant.	06
5(c).	Explain the fuel supply system of a diesel engine power plant.	09
6(a).	With neat sketch, describe the working principle of combined cycle power plant.	15
6(b).	A gas turbine power plant operating on Brayton cycle has a pressure ratio of 8. The gas temperature is 300 K at the compressor inlet and 1300 K at the turbine inlet. Assuming a compressor efficiency of 80% and a turbine efficiency of 85%, determine (i) the gas temperature at the exits of the compressor and the turbine, (ii) the back work ratio, (iii) the thermal efficiency, and (iv) the thermal efficiency of the gas turbine if a regenerator having an effectiveness of 80% is installed	15
6(c).	What are the properties to be considered while selecting the fuel for a gas turbine power plant?	05
7(a).	What are the purposes of intercooling and reheating in gas turbine power plant? Describe.	15
7(b).	Draw a layout of steam power plant with a brief description.	10
7(c).	What activities should be carried out for designing a steam power plant? Describe.	10
8(a).	What are the methods used for feed water treatment? Describe the chemical treatment method with neat sketch.	15
8(b).	Write down the purpose of chimney. Derive an expression for the height of chimney.	10
8(c).	Water at 30°C flows into a cooling tower at the rate of 1.25 kg per kg of air. Air enters the cooling tower at the dry bulb temperature of 20°C and a relative humidity of 60% and leaves it at a dry bulb temperature of 28°C and 90% relative humidity. Make-up water is supplied at 20°C. Determine (i) the temperature of water leaving the cooling tower, (ii) the fraction of water evaporated, and (iii) the approach and range of the cooling tower.	10

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2021

ESE 4105

(Fuel Combustion and IC Engines)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|--|----|
| 1(a). | What is meant by “Alternative fuels”? Discuss the prospects of ethanol and biodiesel as a fuel for IC engines. | 13 |
| 1(b). | What is octane and cetane number? Explain what it means by a cetane rating of 60 and an octane rating of 85? | 10 |
| 1(c). | Explain the differences between:
i. Ultimate analysis and proximate analysis
ii. Higher calorific value and lower calorific value. | 12 |
| 2(a). | Why cooling and lubrication systems are necessary for IC engines? | 06 |
| 2(b). | How lubricating oils are classified? Discuss the mist lubrication system with diagram. | 14 |
| 2(c). | Why petroleum refining is necessary? Explain fractional distillation and reforming process. | 15 |
| 3(a). | Describe the water cooling system of IC engine with a schematic diagram. | 15 |
| 3(b). | Describe the different types of injection nozzles with their relative merits and demerits. | 12 |
| 3(c). | Explain the variation in mixture quality required by a SI engine running with different operating conditions. | 08 |
| 4(a). | How temperature and fuel equivalence ratio affect the conversion efficiency of a catalytic converter? Explain with necessary graphs. | 10 |
| 4(b). | Compare the emissions from SI engine with that from CI engine. | 12 |
| 4(c). | What are the common emission control techniques? Discuss the strategies to reduce NO _x emission from IC engines? | 13 |

SECTION – B

- 5(a). What is meant by “Equivalence ratio”? “Generally a CI engine operates with lean mixture”- Explain. 09
- 5(b). What is the physical significance of volumetric efficiency? “Mean piston speed ranges within 8 to 15 ms⁻¹”- Discuss. 10
- 5(c). With the help of indicator diagram, explain: 10
i. Indicated work/cycle;
ii. Pumping work/cycle.
- 5(d). Why pumping loss is minimum for engine operating at WOT condition? 06
- 6(a). What is meant by scavenging? Why scavenging is more important for 2-stroke engine than for 4-stroke engine. 07
- 6(b). Discuss why actual valve timing differ from theoretical valve timing. Also, write the consequences of valve overlapping. 12
- 6(c). How inlet valve opening and closing, and engine speed affect volumetric efficiency? 09
- 6(d). What are the benefits of supercharging? Why 2-stroke CI engine is more common than 2-stroke SI engine? 07
- 7(a). What are meant by:- (i) surface ignition, (ii) end gas, and (iii) flame front? 11
“Pre-ignition is more harmful than post ignition”- Explain.
- 7(b). Discuss the different stages of combustion in SI engine using appropriate diagrams. 12
- 7(c). What is meant by “Knock” in relation to SI engine? “SI engine performance is knock limited but CI engine not”- justify the statement. 12
- 8(a). How knock in SI engine differs from knock in CI engine? 10
- 8(b). “A good SI engine fuel is a bad CI engine fuel”- justify the statement. 10
- 8(c). Discuss the factors that affect ignition delay in CI engines. 10
- 8(d). Why low ignition quality fuel can be used in very large, slow-speed CI engine? 05

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 4th Year 1st Term Examination, 2021

ESE 4125

(Safety and Environmental Aspects of Energy Projects)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|---|----|
| 1(a). | What are the components of OSHM? Which areas do they address? | 05 |
| 1(b). | Define Hazard. What are the typical sources of Hazard? What are the consequences of Hazard? | 10 |
| 1(c). | How could you assess the impacts of Hazard with the help of a Hazard Matrix? | 08 |
| 1(d). | Write down the type of Hazards imposed by Radiation. Are there any limits for Radiation Exposure? What are the preventive measures? | 12 |
| 2(a). | Write short notes on the following-
i) Emergency lighting, ii) Heat Cramps, iii) Cave in, iv) Asphyxiation | 10 |
| 2(b). | What do you understand by improper housekeeping? What types of accidents can occur due to improper housekeeping? | 07 |
| 2(c). | Explain the three basic relationships between Man, Machine and Environment. What are the remedial measures to ensure safety? | 10 |
| 2(d). | Define BLEVE. Why prevention of BLEVE is necessary? | 08 |
| 3(a). | “All combustible materials are not fuel” -Explain. | 07 |
| 3(b). | What are the fire fighting materials used in the industry? Can water be used to extinguish all types of fire? Explain. | 10 |
| 3(c). | “Event tree analysis is the logical counter part of the fault tree”- Explain. | 06 |
| 3(d). | What are meant by PASS and Fire Triangle? Explain the fire management in High-Rise buildings. | 12 |
| 4(a). | What is meant by PPE? Briefly write about respiratory protective equipment. | 10 |
| 4(b). | What are the different types of safety audits? Explain the important aspects to be kept during execution of safety audit. | 09 |
| 4(c). | Define the followings:
i) Accident frequency rate , ii) Accident severity rate and iii) Fatal accident rate | 06 |
| 4(d). | An undertaking with 600 workers, working 50 weeks a year with weekly working hours of 40 hours has 60 accidents causing, the workers were absent during 5% of the aggregate working time. Find the accident frequency rate. | 10 |

SECTION – B

- 5(a). Write short notes on the followings: 09
i) Occupational health, ii) Occupational Safety and Health Act, and iii) Occupational Safety and Health Administration.
- 5(b). Provide two definitions of occupational disease. Write down the occupational causes of 11
following diseases:
i) Stress reactions ii) Lung diseases, and iii) Blood disorders.
- 5(c). Write down the differences between TLVs and PELs. Explain the different ways of 10
calculating the TLVs.
- 5(d). Briefly discuss about occupational ergonomics. 05
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- 6(a). What is meant by Hazardous chemicals? What are the responsibilities of a supplier and 08
an employer related to MSDS?
- 6(b). Explain the hazardous chemical dose-response relationship. Also point out the following 08
in this relationship:-
i) Lethal, ii) Sub-lethal, and iii) Acute.
- 6(c). What is meant by MSDS? Discuss about the following headings regarding to prepare a 11
MSDS:
i) Identification of substance and company, ii) Hazards identification, iii) First aid measures, iv) Handling and storage, and v) Regulatory information.
- 6(d). Write down the occupational effects of the following types of chemicals: 08
i) Toxic chemical, ii) Corrosive chemical, iii) Reactive chemical, and iv) Flammable chemical
-
- 7(a). Classify and explain the pollutants. Discuss about major sources of air pollution. 10
- 7(b). What are the local and regional environmental impacts posed by thermal power plants 05
and petroleum industries?
- 7(c). What is greenhouse effect? How does CO_2 gas in air increases the global warming? 10
- 7(d). Write down the causes and effects of (i) Acid rain and (ii) SMOG 10
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- 8(a). Why sustainable development calls for pollution prevention and clean technology? What 08
are the environmental challenges of sustainable development?
- 8(b). What are the differences between Environmental Impact Assessment (EIA) and Initial 12
Environmental Examination (IEE)? With the help of a flow chart, explain the EIA process.
- 8(c). Why Environmental Management Plan is essential during a project implementation? 10
Discuss about general process of gaining environmental clearance of a new project.
- 8(d). Briefly discuss about ISO 14001. 05

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